

Secret Sharing Schemes

Lecture 20 — Shamir, CRT, and Brickell

SEC 595 — Encrypted Computing

- 1 Shamir's Scheme
- 2 CRT-Based Secret Sharing
- 3 Brickell's Scheme
- 4 Security and Comparison

Shamir

A (t, n) -threshold scheme based on Lagrange interpolation (Lecture 2).

Interpolation

Given t points $(x_1, y_1), \dots, (x_t, y_t)$ with distinct x_i in a field, there is exactly one polynomial of degree at most $t - 1$ through them.

The secret is the constant term of a polynomial; the shares are evaluations; reconstruction interpolates back to the constant term.

The modulus p is prime, so $\mathbb{F}_p$ is a field and uniqueness holds.

The dealer forms, modulo a prime p ,

$$q(x) = a_0 + a_1x + \cdots + a_{t-1}x^{t-1}, \quad a_0 = S,$$

with $a_1, \dots, a_{t-1}$ uniform in $[0, p)$.

Shares

$$S_i = q(i) \bmod p, \quad i = 1, \dots, n.$$

Participant P_i receives the pair (S_i, i) .

The random coefficients are the dealer's random string: $t - 1$ of them, one fewer than the threshold.

The node $x = 0$ is never handed out: $q(0) = S$.

Example — Shamir distribution

$S = 13$, $n = 5$, $t = 3$, $p = 17$, random coefficients $a_1 = 10$, $a_2 = 2$.

- a. Write down $q(x)$.
- b. Compute the shares $S_1, \dots, S_5$.
- c. What is each participant given?

Any t participants pool their pairs (y_j, x_j) and interpolate at 0:

Recovery

$$S = q(0) = \sum_{j=1}^t y_j L_j \bmod p, \quad L_j = \prod_{m \neq j} \frac{0 - x_m}{x_j - x_m}.$$

$L_j = \ell_j(0)$, the Lagrange basis polynomial evaluated at 0.

L_j depends only on the nodes — on *who* turned up, not on what they hold. It can be computed before any share is revealed.

Example — Shamir recovery

Shares from the previous example, modulo 17.

- a. Participants pool $(y, x) = (7, 2), (10, 3), (0, 4)$. Compute L_1, L_2, L_3 and recover S .
- b. Recover S again from the shares at nodes 1, 4, 5.
- c. Why are the two computations obliged to agree?

Claim

Any $t - 1$ shares leave every candidate secret equally likely.

For each candidate s there is exactly one polynomial of degree $t - 1$ through the $t - 1$ known points and $(0, s)$ — that is, exactly one choice of the random coefficients.

So all p secrets remain consistent, each in exactly one way. Privacy holds exactly, not approximately.

Each share is one element of $\mathbb{F}_p$, the same size as the secret.

CRT-based sharing (Mignotte)

Replace “evaluate a polynomial at n points” with “reduce an integer modulo n numbers”.

Shares

Choose public, pairwise coprime moduli $d_1 < d_2 < \dots < d_n$.

$$S_i = S \bmod d_i,$$

and P_i receives (S_i, d_i) .

Reconstruction is the CRT (Lecture 1): any t residues determine S modulo the product of their moduli.

Condition

$$\underbrace{d_1 d_2 \cdots d_t}_{t \text{ smallest}} > S > \underbrace{d_{n-t+2} \cdots d_n}_{t-1 \text{ largest}}$$

- **Left — correctness.** Any t moduli have product $> S$, so the CRT residue below that product is S itself.
- **Right — privacy.** Any $t - 1$ moduli have product $< S$, so they pin down only a residue class, not S .

The threshold is enforced entirely by the sizes of the moduli.

Example — CRT sharing

$S = 117$, $n = 5$, $t = 3$, moduli $d = 4, 5, 7, 9, 11$.

- a. Check the size condition.
- b. Compute the shares $S_i = S \bmod d_i$.
- c. Recover S from $(1, 4)$, $(2, 5)$, $(7, 11)$ using the idempotents of Lecture 1. Why is the answer unique?
- d. For $n = 7$, $t = 5$, choose moduli and a secret satisfying $d_1 \cdots d_5 > S > d_6 d_7$.

How private is it?

The shares are a deterministic function of S : there is no random string. The right inequality only guarantees *more than one* candidate.

In the example, $\{P_4, P_5\}$ hold $S \equiv 0 \pmod{9}$, $S \equiv 7 \pmod{11}$, so $S \equiv 18 \pmod{99}$. Below 140 the candidates are 18 and 117; knowing $S > 99$ leaves only 117.

Asmuth-Bloom

Add a public modulus m_0 , draw a random A , and share $y = S + Am_0$, with $m_0 d_{n-t+2} \cdots d_n < d_1 \cdots d_t$. The random A restores the missing randomness.

Shamir and CRT realise **threshold** structures. Brickell's scheme realises a **general** access structure $\mathcal{A}$ (Lecture 19).

Idea

Give each participant a public vector $V_i \in \mathbb{F}_p^d$, chosen so that a set is authorized exactly when its vectors span the target $(1, 0, \dots, 0)$.

- every authorized set must span the target;
- every unauthorized set must **not**.

Constructing the V_i is the whole difficulty.

Scheme

- Secret vector $a = (a_1, \dots, a_d)$ with $a_1 = S$ and $a_2, \dots, a_d$ random in $[0, p)$.
- Shares: $S_i = a \cdot V_i \bmod p$; P_i receives (S_i, i) .

As in Shamir: the secret in one distinguished slot, randomness in the other $d - 1$.

An authorized set finds scalars c_i with $\sum_i c_i V_i = (1, 0, \dots, 0)$ — a linear system in the public vectors alone. Then

$$\sum_i c_i S_i = \sum_i c_i (a \cdot V_i) = a \cdot \left(\sum_i c_i V_i \right) = a \cdot (1, 0, \dots, 0) = a_1 = S.$$

The target selects the coordinate holding the secret and discards the random ones.

Like the L_j in Shamir, the c_i can be computed before any share is revealed.

An instance

$$S = 99, n = 4, d = 3, p = 127.$$

Authorized (minimal) sets: $\{P_1, P_2, P_3\}$ and $\{P_1, P_4\}$.

$$V_1 = (0, 1, 0), \quad V_2 = (1, 0, 1), \quad V_3 = (0, 1, -1), \quad V_4 = (1, 1, 0)$$

$$V_2 + V_3 - V_1 = (1, 0, 0), \quad V_4 - V_1 = (1, 0, 0).$$

Random coordinates: $a_2 = 55, a_3 = 38$, so $a = (99, 55, 38)$.

Example — Brickell by hand

The instance above, modulo 127.

- a. Compute the shares S_1, S_2, S_3, S_4 .
- b. Recover S from $\{P_1, P_2, P_3\}$, then from $\{P_1, P_4\}$.
- c. Show that $(1, 0, 0)$ is not in the span of V_2, V_3 . What does this mean for $\{P_2, P_3\}$?

Information-theoretic security

Holds against an adversary with unlimited computing power.

Shamir's privacy argument is a statement about **degrees**: no adversary, no running time, no hardness assumption.

The encryption schemes of this course are **computational**: Regev and GSW rest on LWE, ElGamal on DDH, Paillier on composite residuosity.

Mignotte's scheme is unconditional but leaks; Asmuth–Bloom repairs this.

The three schemes

	Shamir	CRT	Brickell
Structure	(t, n)	(t, n)	general $\mathcal{A}$
Secret in	$a_0 = q(0)$	S	a_1
Share	$q(i)$	$S \bmod d_i$	$a \cdot V_i$
Recovery	Lagrange at 0	CRT	solve for c_i
Privacy	exact	interval-dependent	exact

In all three, reconstruction is a fixed linear function of the shares, determined by which participants turned up.

- 1 Shamir: $q(x)$ of degree $t - 1$ with $q(0) = S$; shares $(q(i), i)$; recovery is Lagrange interpolation at $x = 0$.
- 2 $t - 1$ Shamir shares leave every secret equally likely: information-theoretic security, no hardness assumption.
- 3 CRT (Mignotte): shares $S \bmod d_i$, threshold from $d_1 \cdots d_t > S > d_{n-t+2} \cdots d_n$; deterministic shares leak, Asmuth–Bloom adds randomness.
- 4 Brickell: public vectors V_i , shares $a \cdot V_i \bmod p$; authorized sets combine to $(1, 0, \dots, 0)$, selecting $a_1 = S$.
- 5 All three put the secret in one slot and randomness (if any) in the rest; reconstruction is linear.